

Charles L. Melvin

Charlottesville, VA

USA

+1 (681) 205-9696

jgm6jy@virginia.edu

www.linkedin.com/in/charleslmelvin

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Education

2022 – 2025 **B.A. Cognitive Science**, *University of Virginia*, Charlottesville, VA
Concentration: *Cognitive Psychology* — Minor: *Astronomy* — Minor: *Computer Science*
Study Abroad: UVA in Valencia, Spain (May–Jul 2024)

Relevant Coursework

Data Structures & Algorithms, Cybersecurity, Cyber Defense, Computer Systems and Organization, Artificial Intelligence, Psychology Research Analysis, Cognitive Psychology, High-Level Cognition, Behavioral Neuroscience, Discrete Mathematics and Theory, Software Development Essentials, Independent Study in Computer Science, Calculus, Symbolic Logic, Complex Problem Solving, Business Spanish, Astrochemistry/Interstellar Medium, Astrophysics/Cosmology, Medical Anthropology, Renewable Energy and Sustainability, Spanish Phonetics, Spanish Grammar and Composition, Spanish Literature, Digital Ethics, Cloud Computing

Research Experience

- 2025 – **Researcher, Floodwatch (UVA–MIRA)**, *University of Virginia*
Present Built agentic reasoning modules for a flood-prediction system. Implemented semantic and tonal analysis of emergency messages and push notifications. Developed offline caching and user feedback loops for crowdsourced weather data. Trained a model to classify and prioritize emergency messages by urgency, relevance, and context, improving response times and reducing cognitive load during flood events. Submitted a 20-page independent research paper detailing methodologies, results, and implications for future work in disaster response and emergency alert systems (Preprinted on arXiv at arXiv:2511.05022 and currently under review for the MDPI Algorithms journal). Contributed to shipping the public-facing PWA (Next.js · TypeScript · tRPC · Prisma · Tailwind · Mapbox GL) at floodwatch.io with Postgres-backed auth, map visualization, and offline-ready client.
- 2024 – **Independent Researcher: ALTER EGO**, *Self-directed*
Present Evolved a personal project into research spanning HCI, human and machine cognition, and software engineering. Designed a browser-native conversational agent with layered memory (working, episodic, associative, affective) to model human cognition. Implemented an inspectable architecture in React/TypeScript with IndexedDB/Dexie persistence, exposing state for reproducible experiments. Developed a rule-based affect pipeline with 28 emotion labels and heuristics for keyword, idiom, punctuation, and context cues, enabling interpretable emotion detection. Authored a research agenda framing ALTER EGO as an in silico participant for cognitive psychology studies (working-memory bounds, forgetting curves, mood-congruent recall).
- 2019 – **Pulsar Search Collaborative**, *Green Bank Observatory*
Present Segmented and analyzed Green Bank Telescope radio data to distinguish interference, noise, and pulsar candidates. Collaborated on improving detection methodologies using signal-processing tools.

Technical Capabilities & Milestones

- Systems Proficient in Windows and Linux (Ubuntu, Kali, Debian, Arch) environments; experience with system configuration, troubleshooting, and performance optimization
- Security Hands-on training in cybersecurity and cyber defense; implemented CSP policies, jailbreak prevention, and secure authentication systems
- Performance Optimized site load to ≤ 650 ms (Cloudflare analytics); reduced bundle size 55% and improved First Contentful Paint from 2.8 s $\rightarrow$ 1.3 s

User Engagement	Built Game Modding Archive serving 730 unique visits/month with 1.2k page views; handled 2.6k HTTP requests in 24 h
Cross-Platform	Developed a PWA with offline capabilities; Lighthouse performance score 92/100
Microservices	Built market infrastructure with 4+ microservices and a unified API layer supporting real-time data processing
AI/ML Integration	Developed a lifelike AI application with mutable system prompts, emotion detection, and multi-persona chat
Research Impact	Analyzed 5+ years of Green Bank Telescope radio data for pulsar detection using advanced signal processing
Automation	Built Discord bots for server management, content moderation, and API integration; developed PowerShell and Bash scripting solutions
Database	Experience with SQL, SQLite, PostgreSQL, and IndexedDB; designed efficient data persistence and retrieval systems
Networking	Implemented offline-first PWAs with service workers; developed mesh-style alert dissemination for low-connectivity environments

Professional Experience

- 2025 – **Founder**, *Triet Freelancing*, Sole Proprietorship
Present Founded a sole proprietorship delivering personalized digital solutions and user-centric applications. Design and deploy customized web applications, automation tools, and technical solutions. Provide IT consulting, system troubleshooting, and digital infrastructure support for clients. Built multiple production websites with performance optimization, security hardening, and responsive design.
- 2025 **Teaching Assistant (CS3140: Software Development Essentials)**, *University of Virginia*
Supported instruction, grading, and student assistance for a foundational software development course. Proctored exams and held office hours to help students with coursework and programming challenges. Provided technical troubleshooting and programming guidance.
- 2025 **Customer Service/Sales Associate**, *Variety Wholesalers*
Delivered customer service and support in a fast-paced retail environment. Assisted customers with product selection and inquiries to ensure a positive experience. Managed multiple concurrent requests while maintaining service quality.

Independent Projects

AI & Cognitive Systems

- 2024 – **ALTER EGO (Desktop)**, *Electron · Python*, Desktop proof-of-concept with multi-persona AI chat, unified LLM API (Ollama, OpenAI GPT), emotion detection (roberta-base-go-emotions), ElevenLabs TTS pipeline, JSON/SQLite memory store, one-click dependency wizard. MIT-licensed; deprecated in favor of PWA rewrite.
- 2025 – **ALTER EGO PWA**, *TypeScript · React · Dexie · Workbox · Webpack 5*, Cross-platform rewrite: bundle shrank 55%; FCP 2.8 s → 1.3 s (Lighthouse 92/100). Offline-first Workbox service worker with versioned cache. Client-side vector memory via Dexie/IndexedDB. Performance dashboard (Ctrl + Alt + P) exports JSON (FPS, AI latency, token stats) in developer mode. Selectable language models from OpenAI API. Emulated human cognitive memory architecture: working, episodic, associative, affective, forgetting curve. Emulated emotion detection with 28 labels using keyword/idiom heuristics. Multi-persona support with mutable prompts. Modular voice layer (ElevenLabs TTS + Web Speech). System-prompt verifier blocks jailbreak/injection; CSP in webpack.config.js. MIT-licensed; actively maintained.
- 2025 – **EMA (Expeditionary Modular Agent)**, *FastAPI · React*, Platform for generating task-driven agents with schedulers, active learning, and real-time chat control. Natural-language task interpreter, APScheduler timing, SQLite/Postgres persistence, WebSocket monitoring. Exportable trained agents. MIT-licensed; early development.

2024 – **CARNIS**, *Python · FAISS · Transformers*, Modular web-content pipeline inspired by bioneurological systems. Six-layer architecture: crawling, processing, graphing, generating, insight, integration. Uses Retrieval-Augmented Generation (RAG) with Hugging Face models, FAISS vector DB, and Transformers. Provides CLI tools and web API. Modular by design for component customization. MIT-licensed; active development.

Systems & Infrastructure

2024 – **AWARE System & PriorityFresh Caching**, *Python · Simulation Framework*, Designed and implemented an offline-first alert dissemination system for crisis environments using a novel priority-based cache replacement policy (PriorityFresh). Developed a full simulation pipeline evaluating redundancy, actionability-first ratios, and timeliness consistency under constrained connectivity. Benchmarked PriorityFresh against related policies, demonstrating improved performance, cost viability, and energy efficiency in low-bandwidth, high-urgency settings.

2025 – **MockOS**, *C++ · x86-64 Architecture*, Basic emulated esoteric operating system written in C++ independent of host OS. x86-64 CPU emulation with virtual memory, paging, and segmentation. Kernel core with process scheduling, context switching, and system call planning. Virtual file system plus simulated device drivers. Terminal UI with debugger/profiler tooling. Educational OS sandbox. MIT-licensed; actively developed.

2025 – **Stackbuster**, *LLM-powered*, LLM-powered coding roguelite with 140 procedurally generated stages, difficulty spikes, boss mini-projects, and optional time-attack mode. Quicksave system enforces start-over on failure.

Compilers & Languages

2025 – **Brainfuck Compiler**, *C++*, Brainfuck → C++ toolchain with one-command executable generation, cross-platform support, auto-compiler detection (GCC, Clang, MSVC), and optimization passes for faster output binaries.

Web Development

2022 – **Personal Portfolio Site**, *HTML5 · CSS/SCSS · JavaScript*, Built from scratch at clmelvin2004.github.io/CharlesLMelvin. Deployed on GitHub Pages.

2022 – **Game Modding Archive**, *Cloudflare Pages*, Founded hobby archive at ddmcc.site. Static JSON index + Cloudflare Pages; 730 visits and 1.2k page views/month at 0.6 s median load.

2022 – **UVA Club Running Site**, *HTML5 · CSS/SCSS · Gulpfile · JavaScript*, Maintained official site at virginia.clubrunning.org.

2022 – **Societal AI @ UVA Official Website**, *HTML5 · CSS/SCSS · JavaScript*, Built from scratch at saiuva.com. Deployed on GitHub Pages.

2025 – **Triet Freelancing Market Website**, *HTML5 · CSS/SCSS · JavaScript · EmailJS · Stripe*, Professional site at trietdev.com. Deployed on Vercel.

2025 – **Tune Eternal Hub**, *HTML5 · SCSS · JavaScript*, Static musician landing site with SCSS architecture, live-reload dev server, and componentized HTML.

2025 – **JQilin Hub**, *HTML5 · SCSS · JavaScript*, Static content creator landing site with SCSS architecture and componentized HTML.

2025 – **Trevor DeWall Personal Site**, *HTML5 · CSS · JavaScript*, Multi-page static site with reusable header/footer components and guided client editing docs.

Bots & Automation

2022 – **Discord Bots**, *Python · JavaScript*, Conversational bots with NLP integration. Utility bot for server management and Reddit API mirroring. GitHub repository update-monitor bot (discontinued). "Fairy" moderation bot for Societal AI Discord: auto-filters offensive content, tracks warnings with 3-strike auto-mute, manages roles/permissions, and ships playful commands (Bad Apple, useless web links).

Games & Game Development

- 2024 – **Lehran Engine**, *C++17 · SDL2 · nlohmann/json*, Runtime for JSON-driven game builds.
Present SDL2/SDL2_ttf + nlohmann/json stack with CMake toolchain. Packages manifest, story, gameplay, and timeline data into distributable runtime.
- 2022 – **Ren'Py Mod Template**, *Ren'Py*, Cross-compatible (Windows · macOS · Linux · Android) starter kit.
Present Ships bare-bones core + optional add-ons to reduce bloat. Fully documented. Improves performance compared to contemporary templates by over 40% on diverse hardware.
- 2022 – **Unity & Godot Projects**, *C# · Lua*, Unity (C# and Lua) mini-projects and Godot light prototypes.
Present

Leadership & Community Involvement

- 2025–Present **Technical Lead, Executive Board**, *Societal AI at UVA*
Co-founded a student organization promoting education, literacy, and collaboration in AI/ML. Steer organizational direction and community outreach. Build and maintain the club's official website and digital infrastructure. Organize technical workshops and guest-speaker events.
- 2025–Present **Research Team Leader**, *Floodwatch (UVA-MIRA)*
Currently leading the complete Floodwatch research team, involving teams researching machine learning, fullstack development, and Internet of Things. Manage machine learning team with weekly coordination meetings. Oversee technical implementation, code review, and system integration for flood prediction infrastructure.

Certifications

- 2019 – Green Bank Observatory Certification
Present
- 2024 – Gilman Program Alumni
Present

Awards & Honors

- 2025 NIRCA Cross Country National Runner-up (Team)
- 2024 Gilman Scholarship Recipient
- 2024 NIRCA Half-Marathon National Champion (Team)
- 2024–2025 Two-time NIRCA All-American (Half-Marathon)
- 2021 Harvard Book Award

Technical Skills

- Programming Languages Python, JavaScript, TypeScript, Java, C, C++, C#, Lua, R, SQL, PHP, Bash, PowerShell, ARM/x86 Assembly
- Markup & Styling HTML, CSS/SCSS/Sass, Tailwind, Markdown, JSON, YAML, LaTeX, BibTeX
- Frameworks & Libraries React, Next.js, Nuxt.js, Vue, PyQt5, Electron, FastAPI, Workbox, Webpack 5, Styled-Components
- Databases & Storage SQL, SQLite, PostgreSQL, IndexedDB/Dexie
- AI/ML Tools HuggingFace, Ollama, LLaMA models, OpenAI API, Whisper, ElevenLabs, Azure, Stable Diffusion, Automatic1111, ComfyUI, FAISS, Transformers
- Development Tools Git, Docker, Ren'Py, Unity, Godot, SDL2, CMake
- Cloud & DevOps Azure, Vercel, Cloudflare Pages, GitHub Pages, GitHub Actions, CI/CD
- Operating Systems Windows, Linux (Ubuntu, Kali, Debian, Arch)

Soft Skills

Core Competencies	Rapid learning, effective problem solving, pressure tolerance, adaptability, data analysis & logical reasoning
Additional Skills	Braille literacy, video creation/editing, encryption/decryption concepts, game modding/development

Languages

Spoken	English (Native), Spanish (Fluent), Catalan (Elementary), Japanese (Elementary)
Tactile	Braille (Literate)

Athletics & Extracurricular Activities

2022 –	UVA Club Running
Present	National Champion and two-time All-American men's team (NIRCA Half-Marathon).
2016 –	Competitive Distance Running
Present	Competitive Cross Country, Track, and Marathon athlete.